ETALE RING MAPS PRESERVE THE DIMENSION OF LOCALISATIONS AT CORRESPONDING PRIMES

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Flatness gives going down, hence dimension of the source localisation is at most that of the target. Algebra, Lemma \ref{algebra-lemma-dimension-going-up}, cited at proof lines 10839-10841. Gives the first inequality.
- An etale ring map is quasi-finite and a quasi-finite map has discrete fibres. Algebra, Lemma \ref{algebra-lemma-etale-quasi-finite}, cited at proof lines 10847-10850. Makes the induced chain of primes in the source localisation strictly increasing, giving the second inequality.

Source reference. Original lemma 07QP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9839–9839. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 00OH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 895–895. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00U5: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1099–1099. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *If $A \to B$ is an étale ring map and $\mathfrak{q}$ is a prime of B lying over $\mathfrak{p} \subset A$, then* $\dim(A_{\mathfrak{p}}) = \dim(B_{\mathfrak{q}})$.

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Proof. Namely, because $A_{\mathfrak{p}} \to B_{\mathfrak{q}}$ is flat we have going down, and hence the inequality $\dim(A_{\mathfrak{p}}) \leq \dim(B_{\mathfrak{q}})$, see Algebra, Lemma 00OH. On the other hand, suppose that $\mathfrak{q}_0 \subset \mathfrak{q}_1 \subset \ldots \subset \mathfrak{q}_n$ is a chain of primes in $B_{\mathfrak{q}}$. Then the corresponding sequence of primes $\mathfrak{p}_0 \subset \mathfrak{p}_1 \subset \ldots \subset \mathfrak{p}_n$ (with $\mathfrak{p}_i = \mathfrak{q}_i \cap A_{\mathfrak{p}}$) is chain also (i.e., no equalities in the sequence) as an étale ring map is quasi-finite (see Algebra, Lemma 00U5) and a quasi-finite ring map induces a map of spectra with discrete fibres (by definition). This means that $\dim(A_{\mathfrak{p}}) \geq \dim(B_{\mathfrak{q}})$ as desired. $\square$